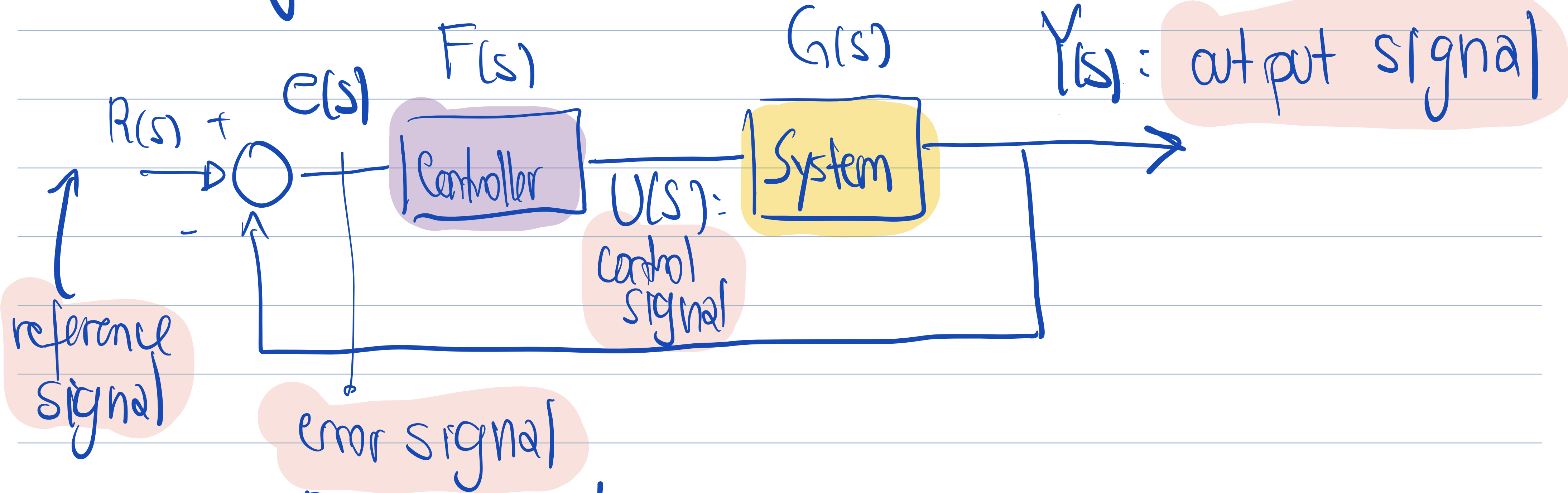
Quick review

Block diagram review:



$E(s) = R(s) - Y(s)$

① OPEN LOOP SYSTEM

G_{OL} : Transfer function between R(s) and Y(s) without feedback

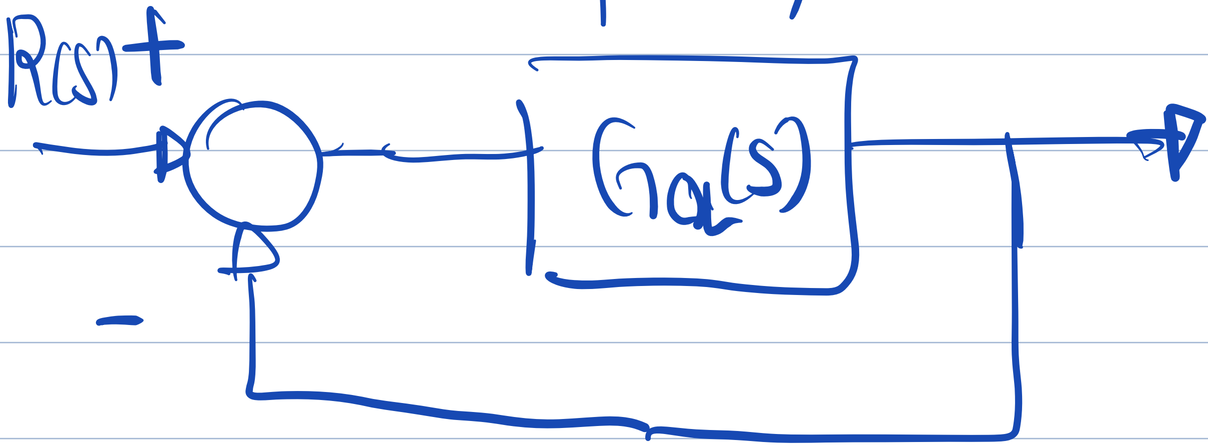
0

Y(s)

$$Y(s) = (R(s) - \cancel{Y(s)}) \cdot F G$$

$$\frac{Y(s)}{R(s)} = F(s)\, G(s) = G_{OL}(s)$$

② Closed Loop system



$$Y(s) = (R(s) - Y(s)) \cdot F \cdot G$$

$$Y(s)(1 + F \cdot G) = F G R$$

$$\frac{Y(s)}{R(s)} = \frac{F(s) \cdot G(s)}{1 + F(s)\, G(s)}$$

$$= G_{CL}(s)$$

$$G_{CL}(s) = \frac{G_{OL}(s)}{1 + G_{OL}(s)}$$

③ PID Controller

A PID Controller is represented as :-

$$F(s) = \underbrace{K_p}_{\text{Proportional term}} + \underbrace{K_I \frac{1}{s}}_{\text{Integral term}} + \underbrace{s \cdot K_d}_{\text{Derivative term}}$$

since $U(s) = F(s) \cdot e(s) = \overbrace{K_p \cdot e(s)}^{P} + \overbrace{K_I \frac{e(s)}{s}}^{I} + \overbrace{s e(s) K_d}^{D}$

If we take the inverse Laplace, we got

TIME DOMAIN:

$$u(t) = K_p\, e(t) + K_I \int_0^t e(t)\,dt + K_d \frac{de(t)}{dt}$$

What this terms represent?

Proportional (P): Represent the present. THE OUTPUT is proportional to the present error
∴ Just a gain

Integral (I): Represent the past. It integrates the past errors. ENSURE 0 steady state error to the reference and to the disturbances

Derivative (D): Represent the future. It Looks at the rate of change of the error (the slope). It improves the stability and the dynamics stability, and the dynamic response of the system (reduce oscillations)

Better damping

→ Improve Damping

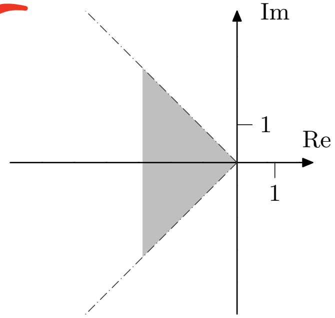


Figure 3.1c

grey area in Figure 3.1c. Poles in this region corresponds to an overshoot of less than 5% (the overshoot may also be affected by the zeros).

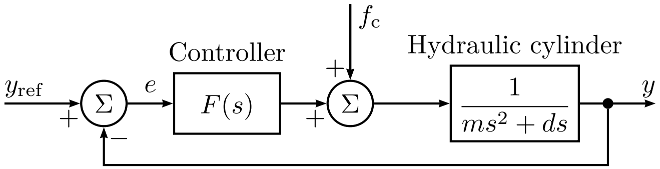


Figure 3.3a

3.3 Figure 3.3a shows the block diagram for a hydraulic servo system in an automatic lathe (Swedish: *svarv*). The output signal $y(t)$ represents the position of the lathe tool (Swedish: *svarvstål*), $y_{ref}(t)$ is the desired position of the lathe tool, m is the mass of the tool slide (Swedish: *verktygsslid*) and the hydraulic piston (Swedish: *kolv*), d is the viscous damping of the tool slide, $F(s)$ is the transfer function for the controller, f_c is the cutting power.

a) How large, in steady state, is the error $e(t)$ between the actual value of the lathe tool and its desired reference value, when there is a step disturbance in the cutting power $f_c(t)$? The controller is assumed to be an amplifier with a constant gain $F(s) = K$.

b) How is this error changed if the amplifier is replaced by a PI controller with transfer function $F(s) = K_1 + K_2/s$?

3.4 Consider the system

$$Y(s) = G(s)U(s) = \frac{0.2}{(s^2+s+1)(s+0.2)}U(s).$$

a) Suppose $G(s)$ is controlled by a proportional controller with gain K_P, that is,

$$U(s) = K_P(R(s) - Y(s)).$$

Use MATLAB to compute the closed loop system, and to plot the step response of the closed loop system. Choose some values for K_P in the range 0.1 to 10. How are the properties of the step response affected by K_P? What happens with the steady state error when K_P increases? Is it possible to obtain a well damped closed loop system and small steady state error using proportional control?

b) Let us now introduce integration in the regulator and use

$$U(s) = (K_P + K_I\frac{1}{s})(R(s) - Y(s)).$$

Put $K_P = 1$ and try some values of K_I in the range $0 < K_I < 2$. How are the step response and the steady state error affected by the introduction of the integrating part and the value of K_I?

c) Finally we will introduce the differentiating part in the regulator and use

$$U(s) = (K_P + K_I\frac{1}{s} + \frac{K_D s}{sT+1})(R(s) - Y(s)).$$

Since true differentiation is difficult to implement, the derivative part is approximated by $\frac{K_D s}{1+sT}$. (This will low-pass filter the error signal before differentiation.) Put $K_P = 1, K_I = 1$ and $T = 0.1$ and try some values of K_D in the range $0 < K_D < 3$. How does the D-part affect the step response of the closed loop system?

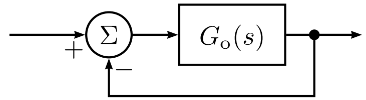


Figure 3.5a

3.5 Draw a root locus with respect to K for the system in Figure 3.5a, with $G_o(s)$ given below. For which values of K are the systems stable? What conclusions on the principal shape of the step response can be drawn from the root locus?

3.1

$$\left[\frac{kg}{s}\right] \frac{d\,m(t)}{d(t)} = \rho \cdot q_{in} - \rho q_{out}$$

$\frac{kg}{m^3}$ $\frac{m^3}{s}$ $\frac{kg}{m^3}$ $\frac{m^3}{s}$

a) " Level control "

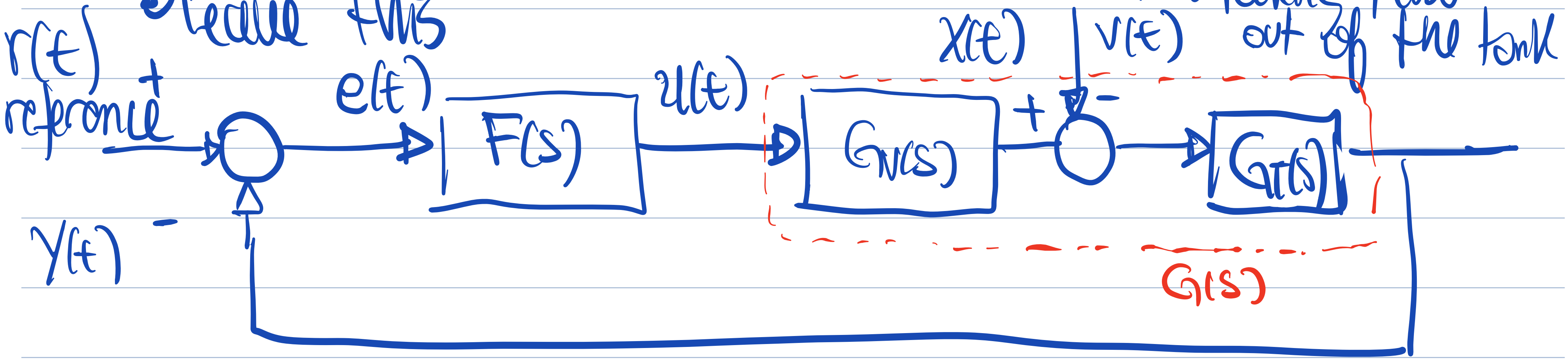
↳ output ($y(t)$) : level

→ input ($u(t)$) : valve position

→ disturbance ($v(t)$) : outflow

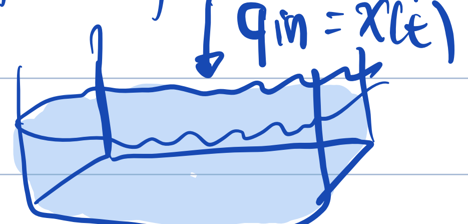
leave this

taking flow out of the tank



Transfer function of the tank $\frac{m^3}{s}$ $\frac{kg}{m^3}$

* →



$$\frac{kg}{s}$$

$$\frac{d\,m(t)}{s} = q_{in}(t) \cdot \rho - q_{out}(t) \cdot \rho$$

$\dot{q}_{out} = v(t)$

dt $\frac{kg}{m^3}$ m m^2 flow rate

we can write $m(t) = \rho \cdot h(t) \cdot A$

$$\cancel{\rho} \cdot \cancel{A} \frac{d\, h(t)}{dt} = q_{in}(t) \cdot \cancel{\rho} - q_{out} \cdot \cancel{\rho}$$

$$\frac{d\, y(t)}{dt} = x(t) - d(t) \qquad / \mathcal{L}$$

$$s\, Y(s) = X(s) - D(s)$$

$$Y(s) = \frac{1}{s} \left(X(s) - D(s) \right)$$

$$G_T(s) = \frac{1}{s}$$

3.24 We don't anything of G(s)

i) P controller
ii) PI "
iii) PD "
iv) PID "

① Integral gain ensure 0 steady state error
So we can take
ii and iv to match C - D
i and iii to match A - B

② Derivative gain anticipate change and reduce oscillations, then we can clarify pair (DAMPINGS)

C with the PID (iv)
D with the PI (ii)

A has better dampings so less oscillation

③ Again the derivative term improves stability

then we can pair

A (which is more stable) with the PD(i,i)

B with P (i)